

HealthPro · AÄ>EB CD>D >C"tO

AôVCDAD4<Cꝱ>C\$8C' BCTECÔVDt=C,,9 Ct2EdB Cô@Câ BCTAD\$CC\$0CÔ=Dđ

Vitest · Vanilla JS ESM · Vite 5 · PWA

AD0D\$0: 03 D\$@C 2CÔO 2026

14

At0CD0Dp

228

B\$5D BEd2

9

BD0C";Ed2

0

At1CäWC

2.4

Bt0D „ACT:)

Bd5C' 4Cä:D4<CT=D" <EdAD\$8D\$L CD5D\$0C'LCÔ8C' >C4;Dô4 C\$8Cꝱ>CÔ0CÔ8DR 7C 2CD0CÔL, C @DTVD\$5CꝱBD.

1. B BC BD4A D4AEdE 14 Ct0C\$4C =DÀ

- #01

A\$ Aꝱ AÔ AÔ

AÄVC4@C FEdO D 5D 5CD>C\$8D"0 Replit

At0C'5Cd=CäAD\$V C\$AD\$0CÔ>C\$;CT=Eb 2 CôVCD:C BC ;Cä7Eb †V ÇF... &ôÔÖö–RÖ!F÷&ðvia/. Workflo
- #02

A\$ Aꝱ AÔ AÔ

A\$8Cô@C 2C'5CÔ=Dò r ?D >C\$0C'LCÔ8DR 4'ÔBCTAD\$VC

health-score.test.js: Cä=Cä2C'5CÔV CäGEd:D42C =Eb 7CÔ0Dt5CÔ=Dò ?Ed4 C'>C4VCꝱC Cô5D 5D >Ct?C
- #03

A\$ Aꝱ AÔ AÔ

A\$8CD0C'5CÔ=Dò EC @CD:Cä4D2 7 toggleHealthTooltip()

At0CÄVD BDÂ @Dô4CꝱVC" PA Ct0CÄVCÔ5CÔ> CÔ0 C\$8Cꝱ;C,,;C, B,v‡2×`eto-crisis'), t('hs-veto-ht2'), t('hs-personal'), t('hs-norm-standard'). AD>CD0CÔ> import { t } CD> health-score.js.
- #04

A\$ Aꝱ AÔ AÔ

AD>CD0CÔ> 6 CÔ>C\$8DR “ †âÔ:C'NDtVC" 4Câ >C >DR AC'>C\$=C,,;Ed2

ui.uk.js D\$0 ui.ru.js: hs-veto-crisis, hs-veto-ht2, hs-veto-hypo, hs-veto-default, hs-norm-personal, hs-norm-standard. B\$0Cꝱ>Cb 4Cä4C =Câ 2Ed4D CD\$=Ed9 data-clear-confirm CD> ui.ru.js.



Aô5D 5C\$VD :C 41 D\$5D B C@CäECä4DôBDÄÄ C VC´4 D4ACôVD,,=C,,9

A 0Ct>C\$8C' =C 1Ed@ D\$5D BEd2 CôVD ;Dò @CTDC :D\$>D 8CÔ3D3 D4AEb C 7CT;CT=Ebà Production

#06

A\$ A AÔ AÔ

B\$5D BC, ?CT@CT<C,,:C =CÔO CÄ>C\$8 A\$ B\$ -CÄVD\$>C¢ ††V ÇF,×66÷&RÖ“ †âçFW7Bæ\$

22 D\$5D BCf Cô5D 5C\$VD :C 7 isis/ht2/hypo D2 V² BC u, norm-mode badge, C´>C4VCª0 show/hide D\$C

#07

A\$ A AÔ AÔ

AT:D ?Cä@D" petBPThresholds() D\$0 getPulseThresholds()

Aä1C,,4C\$V DDCCÔ:DdVEr ?Cä7CÔ0Dt5CÔV export CD;Dò VCT>C´LCä2C =Cä3Cä Væ—BÔBCTAD\$CC\$

#08

A\$ A AÔ AÔ

B\$5D BC, ?Cä@Cä3Ed2 BP D\$0 Pulse (bp-pulse-thresholds.test.js)

57 D\$5D BEd2: CäACä1C,,AD\$0 CÔ>D <C „2D V 5 D <D43), C\$VCD<Cä2C ?D 8 CÔ5Cª>D 5CªBCÔ>CÄ

#09

A\$ A AÔ AÔ

calcHealthScore() Cô>C\$5D BC T { score, status }

AÔ>C\$8C' DCä@CÄ0D" 2Ed4Cô>C\$VCDV: Cä1'ED:D" 7 Dt8D ;Cä2C,,< score (0-100) D\$0 D OCD:Cä2C,,< null). B4AEb ACô>Cd8C\$0DtV Cä=Cä2C´5CÔV: analytics/index.js, export/pdf.js.

#10

A\$ A AÔ AÔ

B CC\$>D 0 Cô5D 5C\$VD :C ?D4;DÄAD3 C\$VCDAD4BCÔV CD0CÔV !' 0 C 0C´VC

scorePulse(null) D\$5Cô5D ?Cä2CT@D\$0EB 7C <EdAD\$L null. AôCC´LD 7C 2Cd4C, 2DT>CD8D\$L CD> C

#11

A\$ A AÔ AÔ

Aª>C´LCä@Cä2C VCÔ4C,,:C FEdO A\$ B\$ D2 T' † æ Àytics/index.js)

Aô@C, FWF –Äæ—5`etoApplied CªVC´LDd5 C 0C´VC" BC =Cä<CT@ Cô@C,,<D4ACä2Cä AD\$0DäBDÄ 6V

#12

A\$ A AÔ AÔ

B\$5D BC, 3D 0CÔ8Dt=C,,E Ct=C GCT=DÄ AT"Aâ †fWFòÖ&÷VæF y.test.js)

41 D\$5D B: sys 179!"180, dia 119!"120 (Cª@C,,7C ´Ä 7—2 S´ C c Ä F— "" C „...@-2), sys 84!"85, dia 54!"55 ×0.60/×0.55. Aô@Ed>D 8D\$5D" 2CTBCâà

#13

A\$ A AÔ AÔ

B\$5D BC, 0CªBC,,2CÔ>D BEb ò :D >Cª>CÄVD 0 (activity-score.test.js)

36 D\$5D BEd2: stepsEnabled=false (Ct=C <CT=CÔ8C£Óf ´Ä R ACÄCC2 ?D >C4@CTAD2 f ó#RóS ósRó

#14

A\$ A AÔ AÔ

B\$5D BC, ?D 8C">CÄC C´VCªVC" † -ÆÇ2×66÷&P.test.js)

31 D\$5D B: 0 Cô@C,,7CÔ0Dt5CÔL (20 C 0C´VC""Ä R ò C´R ò S R ò f´R ò " R² 4CäBD 8CÄ0CÔ=DòÄ 3D 0C C\$8Cª;DäGC TD\$LD O CtV Ct=C <CT=CÔ8Cª0.

2. A @DTVDS5CªBD4@C 4C,,=C <EdGCÔ>C4> Ct=C <CT=CÔ8CªC

B 8D BCT<C >DdVCÔ:C, 7CD>D >C"tO CÔ>D <C ;Ed7D4T D 5CtCC´LD\$0D" =C 1C ;Ed2, C\$@C ECä2D4NDt8 C´8D

2.1 B\$0C ;C,,FDò <Cä4D4;Ed2 D\$0 D4<Cä2 C\$8Cª;DäGCT=CÔO

AÄ>CDCC´L	A\$0C40 (max)	A\$8Cª;DäGC TD\$LD O Cª>C´8
A @D\$5D VC ;DÄ=C,,9 D\$8D : (BP)	40 C 0C´VC	AÔVCª>C´8 — Ct0C\$6CD8 C :D\$8C\$=C,,9
AôCC´LD	20 C 0C´VC	AÔVCª>C´8 (D BD >C40 Cô5D 5C\$VD :C C\$VCDAD4BCÔV CD0CÔV
A´VCª8 (Pills)	20 C 0C´VC	AÔVCª>C´8 — CÔ0C\$VD\$L 0% = 0 C 0C´VC" C Ct=C <CT=CÔ8CªC
@d B" ,,\$Ô´	10 C 0C´VC	height C 1Câ peight CÔ5 Ct0CD0CÔV !' null !' C\$8Cª;DäGC TD\$LD O
A :D\$8C\$=EdAD\$L (Steps)	10 C 0C´VC	stepsEnabled = false !' null !' C\$8Cª;DäGC TD\$LD O

2.2 BD>D <D4;C @Cä7D 0DTCCÔ:D2 66÷&P

```
activeModules = modules.filter(m => m.score !== null)
maxPossible = activeModules.reduce((s,m) => s + m.max, 0)
rawTotal = activeModules.reduce((s,m) => s + m.score, 0)
finalScore = Math.round(rawTotal / maxPossible * 100)
```

2.3 Aª>CÔ:D 5D\$=Eb :Cä=DDVC4CD 0DdVER 7CÔ0CÄ5CÔ=C,,:C

A´8D,,5 BP + Pills	max = 60	AôCC´LD àull (D BC @C ?Cä2CT4Ed=Cª0, CD> v4.1)
BP + Pulse + Pills	max = 80	A 0Ct>C\$0 Cª>CÔDEd3D4@C FEdO (CÔ5CÄ0EB \$Ô´Â 0CªBC,,2
BP + Pulse + Pills + BMI	max = 90	At0CD0CÔ> Ct@EdAD" BC 2C 3D2Â :D >Cª8 C\$8CÄ:CÔ5CÔV
BP + Pulse + Pills + Steps	max = 90	Aª@Cä:C, CC\$VCÄ:CÔ5CÔ>, C ;CR \$Ô´ 2Ed4D CD\$=Ed9
B4AEb R <Cä4D4;Ed2 C :D\$8C\$=E`	max = 100	Aô>C\$=C,,9 Cô@CäDEd;DÄ BMI + Cª@Cä:C, CC\$VCÄ:CÔ5CÔ>

2.4 At<Ed=C ?Cä2CT4Ed=Cª8 CôCC´LD C (v4.1)

Breaking change: scorePulse(null) D\$5Cô5D ?Cä2CT@D\$0EB 7C <EdAD\$L null.

AôCC´LD 7C 2Cd4C, 2DT>CD8D\$L CD> Ct=C <CT=CÔ8Cª0 (max Ct0C\$6CD8 "e 80). UI-Cô@C ?Cä@CTFDÄ VÇ6TW†6ÇVFVB C

3. B ?CTFC,,DEd:C FEdO A\$ B\$ -Cª>CTDEdFEdTCÔBEd2

AäAD\$0CÔ=Ed9 C\$8CÄVD †ÖV 7W&VÖVçG5³ Ô´ ?CT@CT2Ed@DôTD\$LD O CÔ5Ct0C´5Cd=Câ 2Ed4 7-CD5CÔ=Cä3C

AÔ0Ct2C AT"Aà	B4<Cä2C BD 8C45D 0	Aª>CTD.	status Cô>C´5	UI Cª>C´VD
A4VCô5D BCä=EdGCÔ8C´ :D 8Cª	sys "e 180 A Aâ F– &R #	×0.30	crisis	#ef4444
A4VCô5D BCä=EdO 2 D B.	sys "e 160 A Aâ F– &R	×0.60	hypertension-2	#ef4444
A4VCô>D\$>CÔVDð	sys < 85 A Aâ F– Â SP	×0.55	hypotension	#ef4444

3.1 Aô@C,,:C´0CB @Cä7D 0DTCCÔ:D2 ,, :D 8Ct0 sys=180)

```
// A$EEd4CÔV CD0CÔV: sys=180, dia=80, pulse=70, age=50 (Dt>C´>C$VCª
BP score (bad band, 180"d180 && 80"d110) = 5 pts
Pulse score (ok band, 45"d70"d100) = 10 pts
Pills score (CÔ5CÄ0EB ?D 8Ct=C GCT=DÄ´ 0 # G0
rawTotal=35, maxPossible=80 !' pre-veto = round(35/80×100) = 44
Crisis veto (sys"e180): round(44 × 0.30) = round(13.2) = 13
```

4. B 5CtCC´LD\$0D\$8 DDVCÔ0C´LCÔ>C4> CÔ@Cä3Cä=D2 Vitest

228 / 228 D\$5D BEd2 CÔ@Cä9D,,;C, q0

9 D\$5D BCä2C,,E DD0C";Ed2 · 0 Ct1CäWC" r 'C A C\$8Cα>CÔ0CÔ=Dó ~2.4 D

4.1 BD0C";C, BC :Ed;DÄ:EdAD\$L D\$5D BEd2

tests/health-score.test.js	16 D
calcHealthScore(): C 0Ct>C\$V D FCT=C @EdW, A\$ B\$, D BC BD4A, CÔCC´LD Ä ÄÄ", C´VCα8	
tests/bp-pulse-thresholds.test.js	57 D
getBPThresholds() D\$0 getPulseThresholds(): CäACä1C,,AD\$0 CÔ>D <C Ä 2Ed:Cä2Eb 3D CCô8, Ed=D\$5C4@C FEdO	
tests/health-score-i18n.test.js	22 D
toggleHealthTooltip(): A\$ B\$ /CÔ>D <C V² G u, DOM-CÄ>C¢Ä FövvÆR 7–6ÆP	
tests/veto-boundary.test.js	41 D
A4@C =C,,GCÔV Ct=C GCT=CÔO sys/dia CD;Dò 7 isis/HT-2/Hypotension, Cα>CTDEdFEdTCÔBC,Ä ?D VCä@C,,BCTB	
tests/activity-score.test.js	36 D
scoreActivity(): 5 D <D43 CÔ@Cä3D 5D C, stepsEnabled=false, Cα0D BCä<CÔ0 DdVC´L, Ct=C <CT=CÔ8C	
tests/pills-score.test.js	31 D
scorePills(): 0%/50%/90%+, C4@C =C,,GCÔV D\$>Dt:C,Ä Vct>C´ODdVDò 4C B, CÄ>CÔ>D\$>CÔ=EdAD\$L	
tests/bmi.test.js	6 D
calcBMI() D\$0 getBMICategory(): D >Ct@C ED4=Cä:, Cα0D\$5C4>D VErÄ " †à	
tests/bp-norm.test.js	8 D
getBPNorm(): C\$VCα>C\$V D <D43C,Ä >D >C 8D BC =Cä@CÄ0, CÄ>C\$0	
tests/pill-schedule.test.js	11 D
isPillDueToday(): daily, date, weekdays, comma-separated, legacy	

4.2 B >Ct?Cä4Ed; Ct0 Cα0D\$5C4>D VDô<C, BCTAD\$VC

Bä=EdB-D\$5D BC, „GC,,AD\$0 DDCCÔ:DdV	130 (57%)
@d=D\$5C4@C FEd9CÔV D\$5D BC€	67 (29%)
DOM / i18n D\$5D BC€	22 (10%)
A4@C =C,,GCÔV Ct=C GCT=CÔO	9 (4%)

4.3 A\$8Cα>D 8D BC =Eb VCÔAD\$@D4<CT=D\$8 D\$0 D\$5DT=Cä;Cä3EdW

Vitest 4.1.5	Test runner / assertion / mock (vi.mock, vi.mocked)
pdfkit	A45CÔ5D 0DdVDò FDÄ>C4> Ct2EdBD2 ?D >C4@C <CÔ> (Node.js ESM)
Vite 5	Production build — Ct0C\$5D HD4TD\$LD O C 5Cr ?Cä<C,,;Cä:
Vanilla JS ESM	A 5Cr DD 5C"<C\$>D :Ed2 — Dt8D BEb <Cä4D4;EbÄ G&VR×6† ¶–æp
PDFDocument	A4, Helvetica, Cα>C´LCä@Cä2Eb BC 1C´8DdV, D BC BD4A-C 5C"4CdV